

Transcorrelated Jastrow Fitting: Atomization Energies

1 Atomization energies

We compute atomization energies with several xTC method families and benchmark them against SHCI+PBE+CV reference values. Per-basis, per-molecule method discrepancies and the full basis set convergence of error metrics are hosted as interactive plotly figures [1].

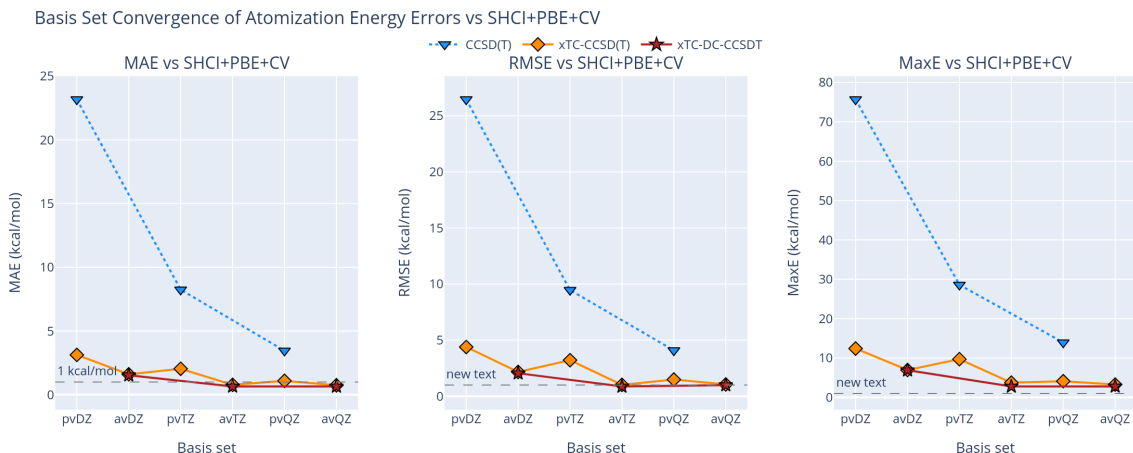


Figure 1: Basis set convergence (MAE, RMSE, MaxE in kcal/mol) of atomization energy errors vs SHCI+PBE+CV for each method family, on a unified x-axis interleaving cc-pVxZ and aug-cc-pVxZ. The interactive version and per-basis per-molecule breakdowns are available online [1].

References

- [1] Kristoffer Simula. Interactive plotly figures: xtc atomization energies vs shci+pbe+cv. Online collection, GitHub Pages, 2026. Landing page: <https://simulak1.github.io/tc-g2-data/>. Basis set convergence of error metrics (MAE, RMSE, MaxE) vs SHCI+PBE+CV: https://simulak1.github.io/tc-g2-data/figures/convergence_vs_shci_pbe.html; https://simulak1.github.io/tc-g2-data/figures/convergence_vs_shci_pbe_QZ_corr.html. Per-basis method discrepancies vs SHCI+PBE+CV: <https://simulak1.github.io/tc-g2->

[data/figures/atomization_method_discrepancy_shci_pbe_pvdz.html](https://simulak1.github.io/tc-g2-data/figures/atomization_method_discrepancy_shci_pbe_pvdz.html); https://simulak1.github.io/tc-g2-data/figures/atomization_method_discrepancy_shci_pbe_pvtz.html; https://simulak1.github.io/tc-g2-data/figures/atomization_method_discrepancy_shci_pbe_pvqz.html; https://simulak1.github.io/tc-g2-data/figures/atomization_method_discrepancy_shci_pbe_avdz.html; https://simulak1.github.io/tc-g2-data/figures/atomization_method_discrepancy_shci_pbe_avtz.html; https://simulak1.github.io/tc-g2-data/figures/atomization_method_discrepancy_shci_pbe_avqz.html; https://simulak1.github.io/tc-g2-data/figures/atomization_method_discrepancy_shci_pbe_avdz_QZ_corr.html; https://simulak1.github.io/tc-g2-data/figures/atomization_method_discrepancy_shci_pbe_avtz_QZ_corr.html.